

```
TARGET = "bank"
SENTENCES = {
    "river": [
        "We sat on the bank of the river and watched the boats.",
        "The children played on the grassy bank near the water.",
        "Heavy rain made the river overflow its bank.",
        "A fisherman stood on the muddy bank all morning.",
        "Ducks were resting on the bank of the lake.",
    ],
    "money": [
        "I deposited my salary at the bank this morning.",
        "The bank approved our loan for the new house.",
        "She works as a manager in a large investment bank.",
        "The bank raised its interest rates again.",
        "He withdrew some cash from the bank before the trip.",
    ],
}
```

=====

STEP 1 - Same word, two contexts: layer 0 (almost static) vs last layer (contextual)

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A: We sat on the bank of the river and watched the boats.

B: I deposited my salary at the bank this morning.

first 5 dims, layer 0 A: [-0.087 -0.911 -1.381 -1.073 -0.433]

first 5 dims, layer 0 B: [-0.021 -0.862 -1.199 -0.961 -0.323]

first 5 dims, last A: [-0.004 -0.42 -0.323 0.112 -0.591]

first 5 dims, last B: [0.517 -0.606 0.153 0.016 0.698]

cosine(A, B) at layer 0 = 0.954 <- same word; only the position embedding differs

cosine(A, B) at last layer = 0.476 <- the context has changed the vector

Note: layer 0 is not exactly 1.0 because BERT adds a position embedding;
a pure static embedding (word2vec) would give exactly 1.0.